CPCTC and Triangle Congruence Proofs

Geometry Worksheet · Grade 9–10

Name: _____ Date: _____

Learning Objectives

- Understand and apply the CPCTC theorem (Corresponding Parts of Congruent Triangles are Congruent)
- Use angle bisector definitions and congruence postulates to write two-column proofs
- Prove that specific angles or sides are congruent using triangle congruence and CPCTC

Problems

1. What does CPCTC stand for?

2. If ray AB bisects angle CAD, what conclusion can you write using the definition of an angle bisector?

$\angle CAB \cong \angle BAD$

3. If ray AB bisects angle CBD, what two congruent angles can you identify?

$\angle CBA \cong \angle ABD$

4. In a two-column proof, if segment AB is shared by triangles CAB and DAB, what statement and reason do you write?

$\overline{AB} \cong \overline{AB}$

5. Given that ∠CAB ≅ ∠BAD, ∠CBA ≅ ∠ABD, and AB ≅ AB, which triangle congruence postulate or theorem proves ΔCAB ≅ ΔDAB?

$\triangle CAB \cong \triangle DAB$

6. Once you have proven ΔCAB ≅ ΔDAB, how can you conclude that angle 1 (∠CAB side) is congruent to angle 2 (∠DAB side) using CPCTC?

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$\angle 1 \cong \angle 2$

7. In the quadrilateral with bisector AB, if CB = 12 and the triangles CAB and DAB are congruent, what is DB?

$DB = 12$

8. Write the correct order of steps for a two-column proof showing $\angle 1 \cong \angle 2$ given that AB bisects $\angle CAD$ and $\angle CBD$: which reason justifies the LAST step?

$\angle 1 \cong \angle 2$

9. Given quadrilateral CADB where AB bisects $\angle CAD$ and $\angle CBD$, write all four steps of the two-column proof proving $\angle CAB \cong \angle DAB$ ($\angle 1 \cong \angle 2$) with correct statements and reasons in order.

$\triangle CAB \cong \triangle DAB \Rightarrow \angle 1 \cong \angle 2$

10. In a different figure, triangles PQR and STU are congruent. If PQ = 3x + 5 and ST = 5x – 7, find the value of x and the length of PQ using CPCTC.

$3x + 5 = 5x - 7$

CPCTC and Triangle Congruence Proofs — Answer Key

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Answer Key

1. Answer: Corresponding Parts of Congruent Triangles are Congruent
- Recall the abbreviation used in geometry proofs.
- CPCTC stands for Corresponding Parts of Congruent Triangles are Congruent.

2. Answer: Angle CAB is congruent to angle BAD
- An angle bisector divides an angle into two equal (congruent) parts.
- Therefore, ∠CAB ≅ ∠BAD by the definition of an angle bisector.

3. Answer: Angle CBA is congruent to angle ABD
- AB bisects ∠CBD means it divides ∠CBD into two equal angles.
- By the definition of an angle bisector, ∠CBA ≅ ∠ABD.

4. Answer: AB ≅ AB; Reflexive Property
- A segment is always congruent to itself.
- Write: AB ≅ AB, Reason: Reflexive Property of Congruence.

5. Answer: ASA (Angle-Side-Angle)
- We have two pairs of congruent angles and the included side AB between them.
- By ASA (Angle-Side-Angle), ΔCAB ≅ ΔDAB.

6. Answer: By CPCTC, corresponding angles of congruent triangles are congruent
- Since ΔCAB ≅ ΔDAB is established, all corresponding parts are congruent.
- Therefore ∠1 ≅ ∠2 by CPCTC.

7. Answer: DB = 12
- Since ΔCAB ≅ ΔDAB, corresponding sides are congruent by CPCTC.
- CB corresponds to DB, so DB = 12.

8. Answer: CPCTC
- The last step concludes that specific angles are congruent after triangle congruence is established.
- The reason for the last step is CPCTC (Corresponding Parts of Congruent Triangles are Congruent).

9. Answer: Step 1: AB bisects ∠CAD (Given); Step 2: ∠CAB≅∠BAD (Def. Angle Bisector); Step 3: AB bisects ∠CBD (Given); Step 4: ∠CBA≅∠ABD (Def. Angle Bisector); Step 5: AB≅AB (Reflexive); Step 6: ΔCAB≅ΔDAB (ASA); Step 7: ∠1≅∠2 (CPCTC)
- List all given information first, then apply definitions and postulates in logical order.
- End with CPCTC after establishing triangle congruence via ASA.

10. Answer: x = 6; PQ = 23

- By CPCTC, corresponding sides of congruent triangles are equal: PQ = ST, so 3x + 5 = 5x – 7.
- Solve: 12 = 2x → x = 6; PQ = 3(6) + 5 = 23.

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